

Project Update

Name: Shahriar Swanon

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Week 1 (22 Jan - 28 Jan):

In the first week, we formed a group of 4 members. We are:

- Shahriar Swanon,
- Md. Abu Sufian Protik,
- Md. Faiaz Bin Hayder,
- Khandakar Atikur Ahsan.

Week 2 (29 Jan – 4 Feb):

In this 2nd week, we selected our project. Our project is **“Create a custom image classification dataset featuring five classes: dog, cow, cat, lamb, and zebra, each with 100 images sourced from the internet or captured using your phone. Develop a classification model to classify these classes with at least 90% accuracy.”**

Week 3 (5 Feb – 11 Feb):

In 3rd week, I researched how to start the project. Since I didn't have proper idea about how to train a model using Google Colab which can classify images. I used YouTube, google and also ChatGPT to learn how to use Google Colab and studied various pre-trained models.

Week 4 (12 Feb – 18 Feb):

In 4th week, I didn't do anything. Because of the National Election and the online classes.

Week 5 (19 Feb – 25 Feb):

In 5th week, I copied a project from YouTube which was a small project about cats and dogs image classifier. I did this to get some ideas about the classifier model. This project helped me to understand the core concepts of the project.

Week 6 (26 Feb – 4 Mar):

In 6th week, I started collecting dataset images. I collected 108 Zebra's images from the internet and renamed those images to (1-108).

Week 7 (5 Mar – 11 Mar):

In 7th week, I collected 111 cat's images for our dataset from the internet and renamed all the images like Cat_(1), Cat_(2)...Cat_(111). During this time rest of my groupmates have already collected other animals photos. Then I uploaded the Animal Dataset to Github repositories.

Week 8 (12 Mar – 15 Mar):

In 8th week, I started building the project. At first, I set up my Google Colab with python 3 runtime and enabled T4 GPU for faster training. Then I stored my Animal Dataset in Google Drive and mounted to the Google Colab environment to access the images during training. Then I cleaned the dataset. Since we collected the images from the internet so, they might contain corrupted or unsupported image formats. That's why I performed a data cleaning step. Each image was opened using the PIL library, converted to RGB format, and saved as a valid JPEG file. Till now I have done this, and I have uploaded the ipynb file to Github repositories.